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LCIA OF IMPACTS ON HUMAN HEALTH AND ECOSYSTEMS

Modeling spatially resolved characterization factors for eutrophication potential in life cycle assessment

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Abstract

Purpose Prior versions of the Tool for Reduction and Assessment of Chemical and other environmental Impacts (TRACI) have recognized the need for spatial variability when characterizing eutrophication. However, the method's underlying environmental models had not been updated to reflect the latest science. This new research provides the ability to differentiate locations with a high level of detail within the USA and provides global values at the country level.

Methods In previous research (Morelli et al. 2018), the authors reviewed a broad range of domain-specific models and life cycle assessment methods for characterization of eutrophication and ranked these by levels of importance to the field and readiness for further development. The current research is rooted in the decision outcome of Morelli et al. (2018) to separate freshwater and marine eutrophication to allow for the most tailored characterization of each category individually. The current research also assumes that freshwater systems are limited by phosphorus and marine systems are limited by nitrogen. Using a combination of spatial modeling methods for soil, air, and water, we calculate midpoint characterization factors for freshwater and marine eutrophication categories and evaluate the results through a US-based case application.

Results and discussion Maps of the nutrient inventories, characterization factors, and overall impacts of the case application illustrate the spatial variation and patterns in the results. The importance of variation in geographic location is demonstrated using nutrient-based activity likelihood categories of agricultural (rural fertilizer), non-agricultural (urban fertilizer), and general (human waste processing). Proximity to large bodies of water, as well as individual hydraulic residence times, was shown to affect the comparative values of characterization factors across the USA.

Conclusions In this paper, we have calculated and applied finely resolved freshwater and marine eutrophication characterization factors for the USA and country-level factors for the rest of the globe. Additional research is needed to provide similarly resolved characterization factors for the entire globe, which would require expansion of publicly available data and further development of applicable fate and transport models. Further scientific advances may also be considered as computing capabilities become more sophisticated and widely accessible.

Keywords Life cycle impact assessment · Characterization · Characterization Factors · Eutrophication potential · Regionalization · Spatialization

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1 Introduction

Eutrophication is a global environmental issue that spans the freshwater-marine continuum (Paerl 2009). Excess nutrients, especially nitrogen (N) and phosphorus (P), enter the biosphere through point and non-point sources of pollution and often contribute to harmful algal blooms (HABs). These nutrients eventually reach surface waters and result in degraded water quality conditions, impacting aquatic ecosystems and human health as well as local economies.

Life cycle assessment (LCA) provides a standardized quantitative framework from which to evaluate

eutrophication potential of freshwater and marine systems (measured in kg N_{eq} and kg P_{eq}, respectively) of a product or process across its lifespan. Life cycle inventory (LCI) models provide the data needed for LCA, while life cycle impact assessment (LCIA) methods translate the data into potential impacts. Various existing LCIA methods incorporate eutrophication models, and a few models exist separately; however, each method and model informs a different aspect of this environmental issue (Morelli et al. 2018). Furthermore, many LCIA models lack the spatial precision needed to operate on a refined global scale, often assigning a single characterization factor (CF) to areas actually representing many different conditions. The data and models required to support this high level of spatial resolution have been developed inconsistently across the globe, leaving major gaps in the research.

Characterization factors for eutrophication potential in the Tool for Reduction and Assessment of Chemical and other environmental Impacts (TRACI) estimate the eutrophication potential of a release of chemicals containing N or P to air or water, per kg of chemical released (Bare 2011; Bare et al. 2002; Norris 2003). Characterization of eutrophication potential in LCA would benefit substantially from improved spatial resolution of the environmental processes, including fate, transport, and potential environmental impacts to air, land, and water (Finnveden and Potting 1999). This paper builds from the eutrophication model review presented in Morelli et al. (2018) and its set of recommendations for improvement to the eutrophication potential category in LCA.

In the current work, we present a first effort to spatially differentiate eutrophication CFs to improve the scientific robustness of fate and transport mechanism characterization. We focus our efforts on the contiguous USA, using TRACI as the LCIA method (Bare 2012), and thereby update TRACI's eutrophication potential impact category to provide a state-of-the-art global site-specific model for research and public use.

2 Methods

2.1 Nutrient framework

In keeping with current LCIA practice, this project uses the simplified assumption that N and P cause eutrophication impacts in systems where they are the limiting nutrient; i.e., freshwater systems are P-limited and marine systems are N-limited. This simplification should be updated in future work, as co-limitation can occur in marine and freshwater systems (Morelli et al. 2018; Payen and Ledgard, 2017; Schindler 2006).

2.1.1 Freshwater eutrophication

The freshwater eutrophication fate model is provided by Helmes et al. (2012), and is driven by phosphorus limitations. This model is based on a 0.5° × 0.5° (approximately 50 km × 50 km at mid-latitudes) hydrologic model and returns characterization factors (CFs) for phosphorus emissions (Fekete et al. 2002). While higher resolution datasets are becoming available (Lehner and Linke 2015), these have not yet been linked to phosphorus fate in freshwater systems. The current fate model accounts for phosphorus removal processes (i.e., advection, retention, and water use) through water withdrawal, uptake by biomass, and adsorption to suspended solids in waterways. It does not account for particulate P in the transfer fraction from agricultural soils to surface freshwater. Nor does it account for sewage treatment removal of P, which can lead to an overestimation of P transport for flows passing through urban centers.

2.1.2 Marine eutrophication

Recent modeling efforts by Cosme et al. (2017b) provide the first LCIA model to address marine eutrophication. This model provides spatially explicit fate factors describing N emissions to soil, freshwater, and marine systems. The CFs are derived from the NEWS2 model (Mayorga et al. 2010), which provides inland soil and freshwater fate for dissolved inorganic N at a watershed scale. In the Cosme et al. work, the residence time of N in the receiving marine systems is modeled as a function of advection and denitrification.

The NEWS2 model, and thus its output, operates at the basin scale, even though some inputs are added at the sub-basin scale and some sub-basin processes are modeled explicitly. While natural and anthropogenic sources of N are distinguished at the sub-basin level, this modeling refinement is only for source contribution analysis in the final model output. In the LCIA application of Cosme et al. (2017b), the NEWS2 soil export fraction is a combination of natural and agricultural soil.

The NEWS2 model does not distinguish between various emission fates within a watershed. For large watersheds, such as the Lower Mississippi River Basin in the USA, there could be significant differences in fate of emissions at the headwaters compared to the outlet of a watershed. Cosme et al. (2017a) and Morelli et al. (2018) identify other mechanistic models for N fate (e.g., AQUATOX, IMAGE-GNM), but both authors note the high data demands and computational intensity of these models across large spatial domains, which make global applications difficult.

In order to translate air emissions to marine impacts, we use the source-receptor model of Roy et al. (2012), which models deposition of NH_x and NO_x to land resulting from air emissions at 2° × 2.5° grid cell resolutions. Roy et al. used GEOS-Chem (Bey et al. 2001), a global 3-D model

for atmospheric chemistry and emission transport, to create these source-receptor matrices based on meteorological data for 2005, which represents average annual temperature and precipitation data from 1961 to 1990.

Given this background, the marine eutrophication model we propose for incorporation into TRACI is the combination of the GEOS-Chem air transport model and the NEWS2 hydrological model, as implemented in Bjørn et al. (Bjørn et al. 2020). Coupling these two models to the NEWS2 basins and marine systems of Cosme et al. makes it possible to trace an emission of N to air, following it to soil, freshwater, and finally to a marine system.

2.2 Fate calculations

Figure 1 summarizes the steps of the fate factor calculation process for both freshwater and marine eutrophication. Since fate represents the midpoint for eutrophication, this process also reflects the calculation of the midpoint characterization factors.

In the LCIA context (Rosenbaum et al. 2007), the fate factor (FF) describes the fate of an emission in a given compartment and location, tracing the emitted substance to its receiving compartments (in the case of multi-media transfer) and locations. In the context of TRACI, the FF is the midpoint characterization factor (CF). Therefore, in this work, FF is used interchangeably with CF. In the case of a single compartment, FF can be expressed as a matrix, as in Eq. (1):

$$\boldsymbol{FF}_{e-r,ij} = \frac{mass\ substance\ in\ compartment\ r,\ location\ j}{(mass\ emitted\ to\ compartment\ e,\ location\ i/year)} \quad (1)$$

Note that FF is often represented in units of time (year, implicitly canceling the masses of Eq. (1)). In the case of a unit emission, FF can describe the steady-state mass in the receiving compartment. In a spatially explicit framework, a single emission location, *i*, may lead to a steady-state mass in one, many, or all other compartments, *j*.

$\mathbf{FF}_{e\text{-}r}$ can be used directly if the compartments and locations modeled match the needs of the analysis. In the case of this work, the following fate factors were used directly:

- freshwater eutrophication due to P emissions to freshwater (Helmes et al. 2012), with a 0.5°×0.5° grid cell resolution
- marine eutrophication in large marine ecosystems (LMEs) due to nitrogen emissions to LMEs, inland freshwater, or inland soil (Cosme et al. 2017a), with an emission resolution of NEWS2 hydrological basins (Mayorga et al. 2010), and an impact resolution of LMEs (Sherman 2001). A large marine ecosystem is a region of ocean 200,000 km^2 or greater, including coastal areas from river basins and estuaries, which is characterized by certain bathymetry (depth), hydrography (physical features), productivity, and trophically dependent populations of a food chain (Sherman 2001).

However, other fate factors require manipulation prior to use in a eutrophication context:

- freshwater eutrophication due to emissions to soil are calculated based on a 10% factor soil-applied nutrients similar to the ReCiPe approach (Huijbregts et al. 2017). This factor is used to facilitate characterization of agricultural data where more site-specific modeling is not possible while also capturing site-specific or watershed-specific P losses from the field in the life cycle inventory stage.
- Marine eutrophication due to nitrogen emissions to air, followed by deposition on LMEs.
- Marine eutrophication due to air emissions of NH_x and NO_x (Roy et al. 2012), with a 2°×2.5° grid cell resolution, followed deposition to freshwater or land, and subsequent transfer to LME,

A marine eutrophication fate factor, $\mathbf{FF}_{\text{NEWS2_comp-LME}}$, relates emissions to a given NEWS2 basin compartment (freshwater, soil, or LME) and location, *i*, to an increased mass of nitrogen in a corresponding LME, *j*. A dimensionless source-receptor matrix, $\mathbf{SRM}_{\text{air-ground}}$, relates emissions to air in location i to a deposition of that substance in location *j*. We use an intermediate matrix, a spatial transform (ST, also dimensionless), to relate the geographic fractions of the cells modeled in the air model to the geographic (and compartment) fraction of the NEWS2 basin. In the matrix $\mathbf{ST}_{\text{air-NEWS2}}$, each element of the matrix describes the fraction of the corresponding air cell (rows) to the corresponding NEWS2 basin's appropriate compartment (freshwater or soil). Therefore, the matrices may be multiplied as shown in Eq. (2).

$$\boldsymbol{FF}_{air-LME} = \boldsymbol{SRM}_{air-ground} \times \boldsymbol{M}_N \times \boldsymbol{ST}_{ground-NEWS2_comp} \times \boldsymbol{FF}_{NEWS2_comp-LME} \quad (2)$$

where the resulting matrix, $\mathbf{FF}_{\text{air-LME}}$, describes the steady-state mass of nitrogen in the LMEs due to an emission to air. Note that there is a stoichiometric conversion, M_N, of the NH_x or NO_x in the SRM model to the nitrogen (N) required by the marine model.

The output of the approach described above includes matrices of the fate data, linking all emission locations and compartments to all receiving locations and compartments. To make this data usable in an LCIA model, it is necessary to aggregate these locations to political boundaries (Sect. 2.3).

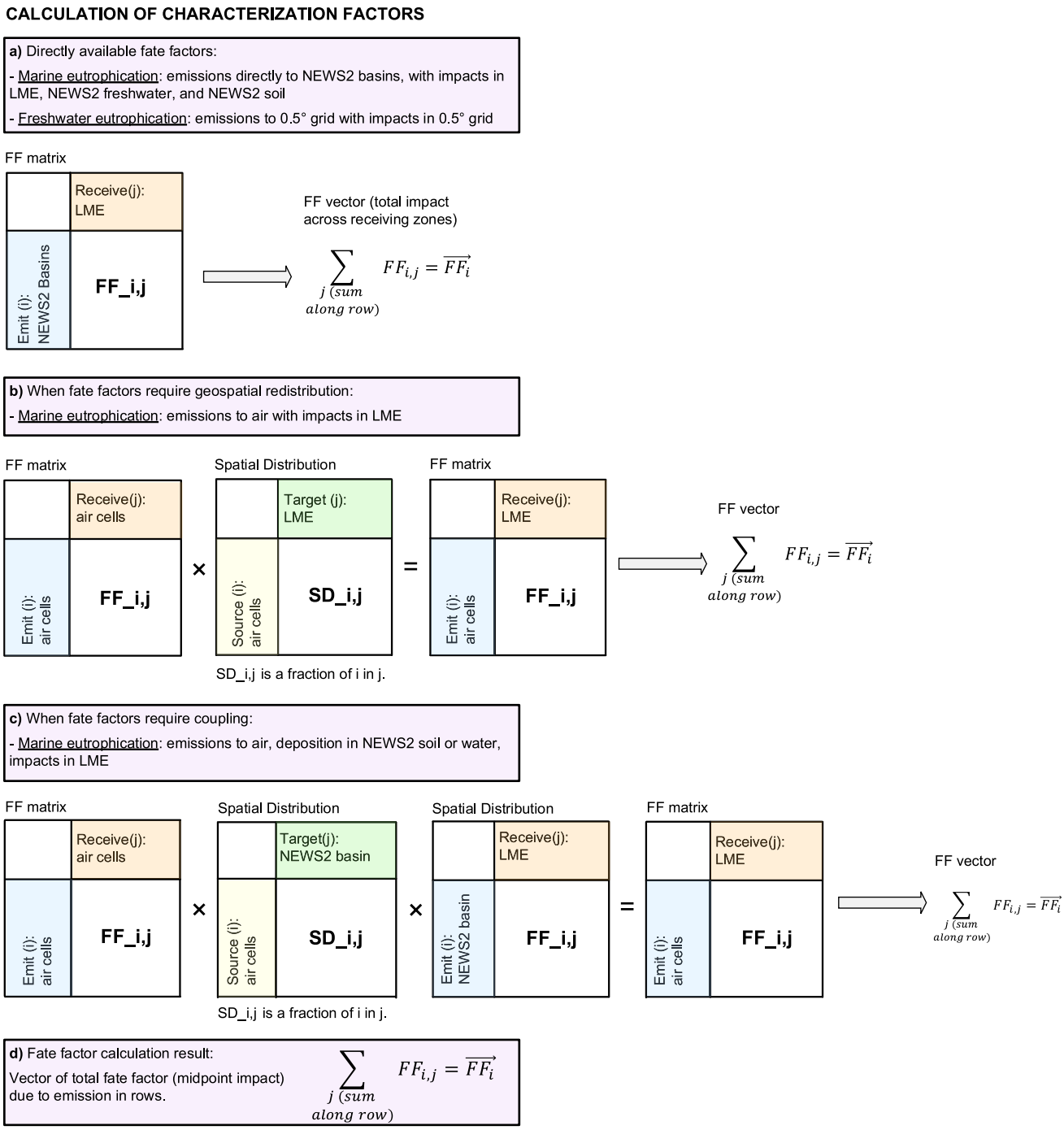

Fig. 1 Calculation of midpoint characterization factors for eutrophication potential. Each step of the process is described in the purple boxes. For each of the matrices, where applicable, the source vector is shown in yellow, the emissions vector is shown in blue, the receiving vector is shown in orange, and the target vector is shown in green

2.3 Aggregation

The goal of geospatial aggregation is to provide representative CF values for a set of target polygons (i.e., polygons to which life cycle inventory data may be connected), based on a set of native polygons (i.e., the original model polygons, such as a grid), using a geographic information system (GIS). In this work, the target polygons are (1) the USA, (2) its states, and (3) its counties. The native CF spatial system is defined by the underlying fate models; for air emissions, the 2°×2.5° grid of Roy et al (2012); for freshwater emissions of P, the 0.5°×0.5° grid of Helmes

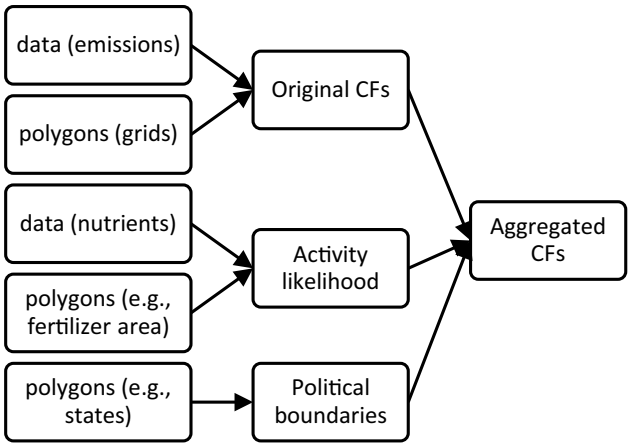

Fig. 2 Schematic of the characterization factor aggregation process. The aggregated CF (rightmost column) is comprised of the original (i.e., "native") CFs, the activity likelihood, and the political boundary (county, state, or country). Each of these elements in the middle column are calculated from corresponding data and/or polygon inputs, as shown in the leftmost column

et al. (2012); for freshwater and soil emission of N, the NEWS2 basins of Cosme et al. (2017a, b); and for emissions of N to the marine system, the LMEs of Cosme et al. (2017a, b). We use activity likelihood data as described below to calculate a representative aggregated value of the target polygon.

The output of the aggregation calculation is a set of CFs for the target polygons (Fig. 2).

The process of aggregating CFs into target units (e.g., states, counties) requires data that represent the intensity of a relevant activity on each native CF (here, a grid cell). The activity likelihood data, or simply "likelihood data," are the "weights" that influence each native CF contribution to the aggregation target. They represent how likely it is that an activity (agricultural, non-agricultural, or unknown/general) occurs on the space represented by the target. For each impact category and substance, there are potentially three sets of likelihood data required. One set is for agricultural activity, which includes applications of fertilizer (or directly modeled emissions from agriculture) as the proxy, or a representative literature value for agricultural activity inside the target polygon. The second set is for nonagricultural activity, which includes industrial emissions, transport emissions, and general population (e.g., human waste) emissions. Finally, an unknown or "general" likelihood dataset is needed for those cases in which the source of an emission is unknown or is being used for the background of an LCI model. The general likelihood dataset should be the sum of the agricultural and nonagricultural likelihood datasets. See the Electronic Supplementary Material (ESM 1) for a summary of phosphorus and nitrogen data used for the likelihood datasets.

When aggregating characterization factors, a few points need to be kept in mind. First, CFs, target, and likelihood polygons need not be co-located, nor do they need to have the same spatial extent. For example, the CF grid does not need to overlap with the political grid, which does not need to overlap with the likelihood grid. Furthermore, the CF grid may extend beyond the borders of the target polygon. Third, likelihood data may be considered extensive or intensive, where extensive values depend on the area considered and intensive values are independent of the area. For example, population (number of people) or fertilizer application (kg N applied) would be extensive values whereas population density (people/km^2) or fertilizer application rate (kg N/ha) would be intensive values. Note that CF values are intensive values. If two equal-area, equally weighted CF polygons are contained in a single target polygon, the resulting target polygon will be the arithmetic average (not the sum) of the two CFs. Finally, there are a few scenarios in which CF and likelihood values are assumed to be zero. When target polygons are missing CF data, the CF is assumed to equal zero. This situation occurs if a target contains areas outside those covered by the CF model. For example, the 0.5°×0.5° freshwater grid does not extend to some coastal border areas of states (i.e., where it is difficult to model hydrologic flow), such as the southern tip of Maryland. In the case of Maryland, since other parts of the state have CF and likelihood data, there will still be a state-level CF value for the state. Another scenario for assumed-zero values is when an area has no likelihood polygons; in this case, the likelihood value is assumed to be zero. This could occur, for example, if a state has areas without fertilizer application data; in this scenario, it would be assumed that no fertilizer is applied in the polygon.

The type of likelihood data also affects the aggregated CF value. For example, consider a target (state) polygon containing several native CF polygons. Suppose that the rural area of the state has high agricultural activity and happens to have high CF values. Suppose that the populous area of the state, with low agricultural activity, tends to have low CF values. An arithmetic average of CFs in the state would yield an overall state CF with a value between the low and high values. However, if agricultural activity is introduced for the likelihood dataset, then an aggregated CF for the entire state would have a higher CF than the arithmetic average because it would be weighted by the higher-value CFs in the agricultural areas of the state.

Figure 3 illustrates the GIS process required to calculate the aggregated CF using the native CF (CF), target (T), and likelihood (L) datasets. Areas *S*, *T*, and *L* can exhibit a variety of relationships with respect to spatial extent and scale. For example, while Roy air cells have global area coverage and tend to be similar in size to counties in the Eastern USA, Roy freshwater grid cells are restricted to continents and tend to be smaller than U.S. counties. It is also possible to have

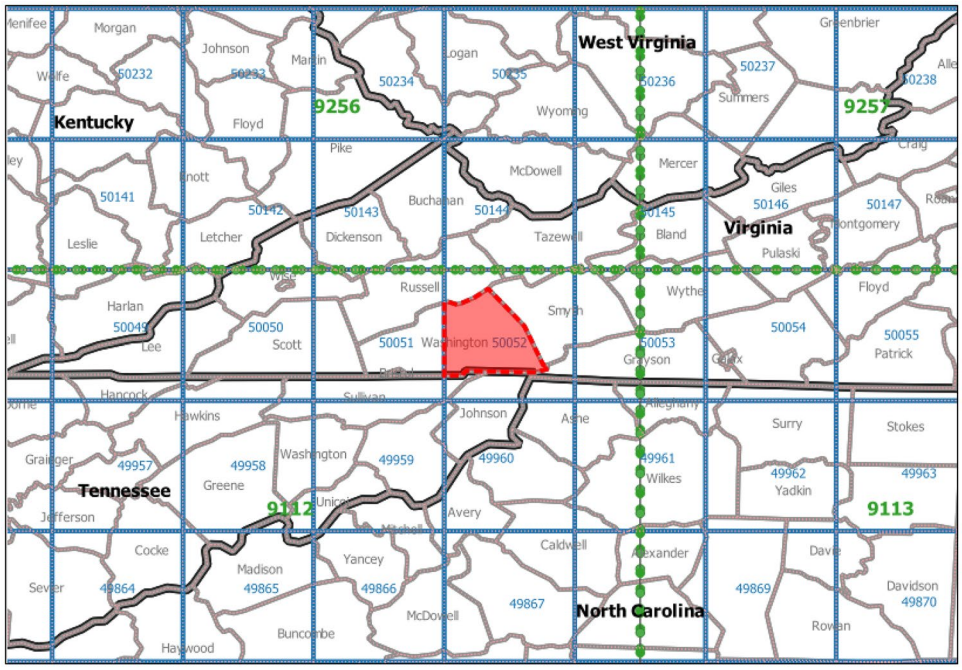

Fig. 3 Spatial intersect of native-scale characterization factors provided by fate models (CFs, green grid), target polygons (*T*, grey lines) for which a representative CF value is desired, and likelihood polygons (*L*, blue lines) with data used to inform the aggregated CF calculation. The intersection polygons are used in the summation of Eqs. (3) and (4). Different likelihood data (e.g., fertilizer application rates, population) may be used to calculate different aggregated CFs

native CFs (i.e., CF data) extend beyond the target polygons, as would be the case with an air emission CF grid in this study.

To perform the CF calculations in a geodatabase, a spatial intersect of CF, T, and L polygons is needed. This results in a table of new polygons, or set *N*, created by the intersect.

Next, we calculate aggregated values for each target zone *t*, according to Eq. (3) for intensive *L* data (already normalized to area) and Eq. (4) for extensive *L* data (which requires dividing *L* data by total L area):

$$CF_{target} = \sum_{n \in t} CF_n * \frac{A_n * L_n}{\sum_{j \in n} A_j * L_j} \tag{3}$$

$$CF_{target} = \sum_{n \in t} CF_n * \frac{A_n * L_n * \left(\frac{A_n}{A_{Ln}}\right)}{\sum_{j \in n} A_j * L_j * \left(\frac{A_j}{A_{Lj}}\right)} \tag{4}$$

where:

n are intersection polygons, members of the target zone *t* (see Fig. 3);

Table 1 Summary of characterization factors calculated in this work. Each row represents a set of CFs, where the set is comprised of the unique combination of each row element listed

Eutrophication category	Substance flow	Emission compartment	Emission resolution	Transfer compartment	Receiving compartment	Receiving resolution	Aggregation resolution	Likelihood data
Freshwater	P	Freshwater	0.5°×0.5° cells	-	Freshwater	0.5°×0.5° cells	County, state, nation	US EPA
Marine	N	LME	NEWS2 Basins	-	LME	LME	County, state, nation	US EPA
Marine	N	Freshwater, Soil	NEWS2 Basins	-	LME	LME	County, state, nation	US EPA
Marine	NH_x, NO_x	Air	2°×2.5° cells	-	LME	LME	County, state, nation	US EPA
Marine	NH_x, NO_x	Air	2°×2.5° cells	Freshwater	LME	LME	County, state, nation	US EPA
Marine	NH_x, NO_x	Air	2°×2.5° cells	Soil	LME	LME	County, state, nation	US EPA

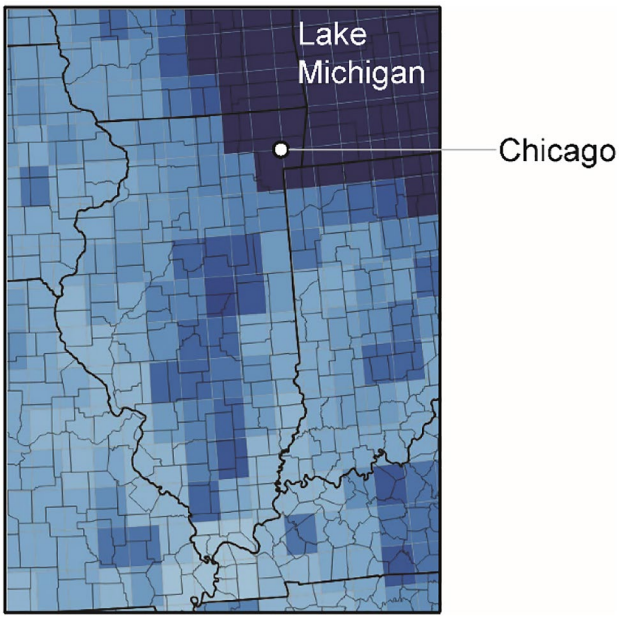

Fig. 4 Maps of characterization factors for freshwater eutrophication to freshwater (top row) and freshwater eutrophication from soil (bottom row). Individual map titles are arranged as [emission compartment]; [likelihood category]. Characterization factor units are years (yr)

CF_n (days) and L_n (e.g., kg fertilizer, for Eq. (4)) are the CF and likelihood data corresponding to polygon n created by the spatial intersect (these data come directly from the corresponding parent polygon, as described in the next paragraph); and

A_x (km^2) is the area of polygon x.

Each polygon N_i inherits information from the parent polygons of which it is a member. In the example in Fig. 3, the red polygon $N_{\mathrm{CF}=9112,\, t=\text{Washington County VA},\, l=50052}$ (here called N_{ex}) represents an area in target $T_{\text{Washington County, VA}}$ where the CF of CF_{9112} and likelihood data L_{50052} apply. N_{ex} has an area smaller than any of its parent polygons, and thus, fractional areas can be calculated as needed. For example, if L holds total fertilizer application (in total kilogram, an extensive variable), then the fertilizer applied in N_{ex} is equal to $A_{N\mathrm{ex}}/A_{L50052,}$ where A represents the area of each. If L holds fertilizer rate (area density in kg/m^2), then the fertilizer rate in the polygon N_{ex} is equal to the fertilizer rate in L_{50052}. A given polygon, N_i, can be larger than some of the parent polygons, but cannot be larger than all three parent polygons, since it was created from a spatial intersect.

2.4 Overall calculations

Each set of aggregated characterization factors for eutrophication potential is specified by the following:

- substance flow (P, N, or N-compound)
- compartments: emission compartment, any intermediate transfer compartment, and receiving compartment
- aggregation resolution
- likelihood data: agricultural, nonagricultural, and general

Table 1 summarizes the characterization factors calculated in this study. Each set of characterization factors is comprised of the aforementioned components for each impact category and aggregated to the county,

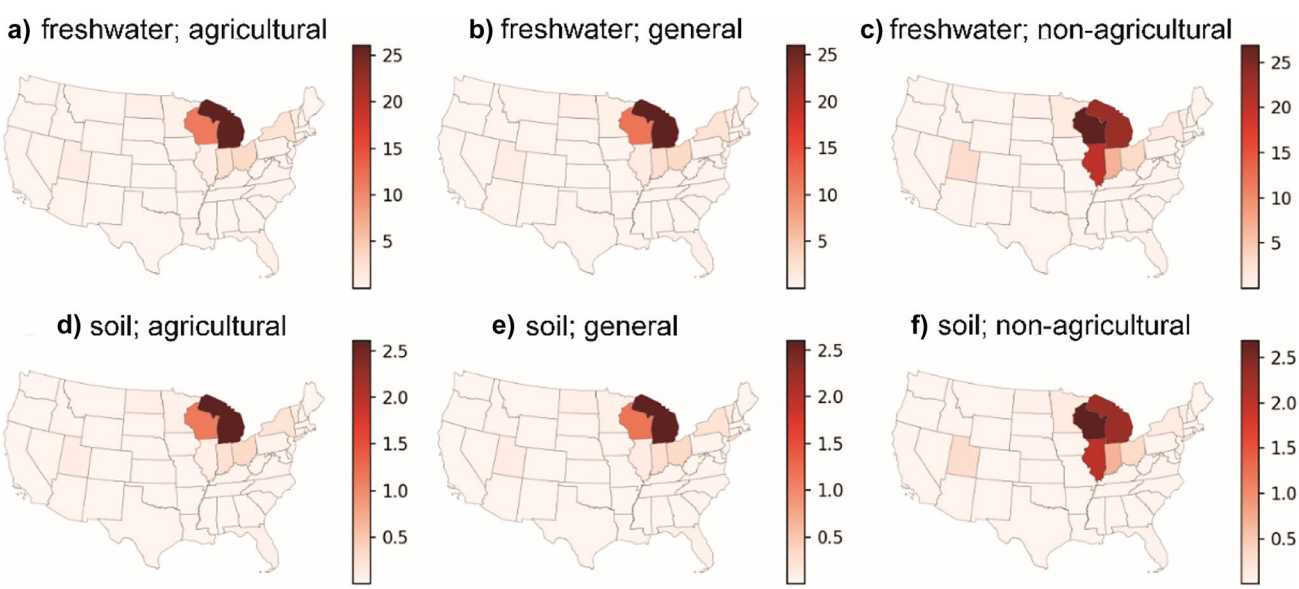

Fig. 5 Maps of characterization factors for marine eutrophication emissions to freshwater, LME, and soil (top row) and air (NH_3 and NO_x; middle and bottom rows). Individual map titles are arranged as [substance emitted]; [emission compartment]; [likelihood category]. Characterization factor units are years (yr)

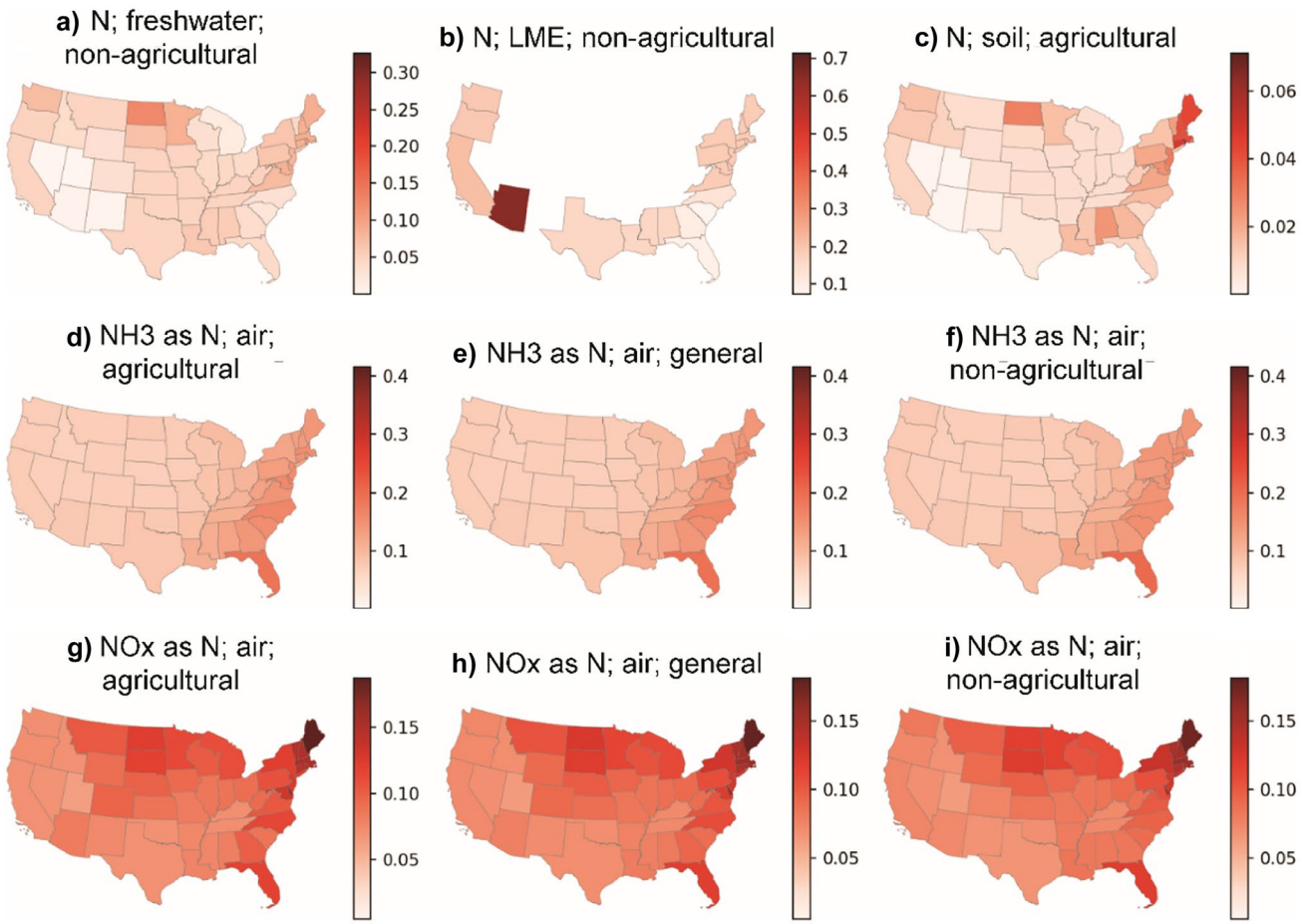

Fig. 6 Application case for P at the state level: agricultural fertilizer (top row), urban fertilizer (middle), and human waste (bottom). Columns represent the national P inventory at the state level (left column, kg P / yr), the corresponding characterization factor (center column, CF, yr), and the total freshwater eutrophication impact (right column, kg P)

state, and national levels. Freshwater eutrophication has one substance (P) and one fate model (P emissions due to freshwater, mapped to $0.5° \times 0.5°$ grid cells). Marine eutrophication has more modeling combinations, as there are three substance flows (N, NH_x, NO_x), four emission compartments (LME, freshwater, soil, and air, with air having two potential transfer compartments, freshwater and soil), and two emission resolutions (NEWS2 basins and $2° \times 2.5°$ grid cells). Altogether, there are 18 sets of CFs for freshwater eutrophication and 27 sets of CFs for marine eutrophication.

3 Results

Freshwater and marine eutrophication CFs can be found in the Electronic Supplementary Material (ESM 2). The data and models used to develop these calculations are available on US EPA's Environmental Dataset Gateway (https://edg.epa.gov) and GitHub project repository (https://github.com/USEPA/lcia-eutrophication). Characterization factors will also be made available on the US Federal LCA Commons (https://www.lcacommons.gov). Maps have been included in this section (Figs. 4, 5, 6, 7 and 8) to facilitate greater understanding of the characterization factors and their spatial patterns.

Table 2 summarizes the statistical dispersion of the CF values, according to the interdecile range. The freshwater and marine eutrophication impact categories are further separated into their emission compartments with corresponding substance flows. These are then each linked to a likelihood category (i.e., agricultural, nonagricultural, or general) and aggregation target (i.e., state or county).

State and county CF values vary based on the spatial nature of the underlying fate models. For example, the freshwater eutrophication model has values that vary by several orders of magnitude. In addition, the model may return a CF value of zero for arid regions of the country.

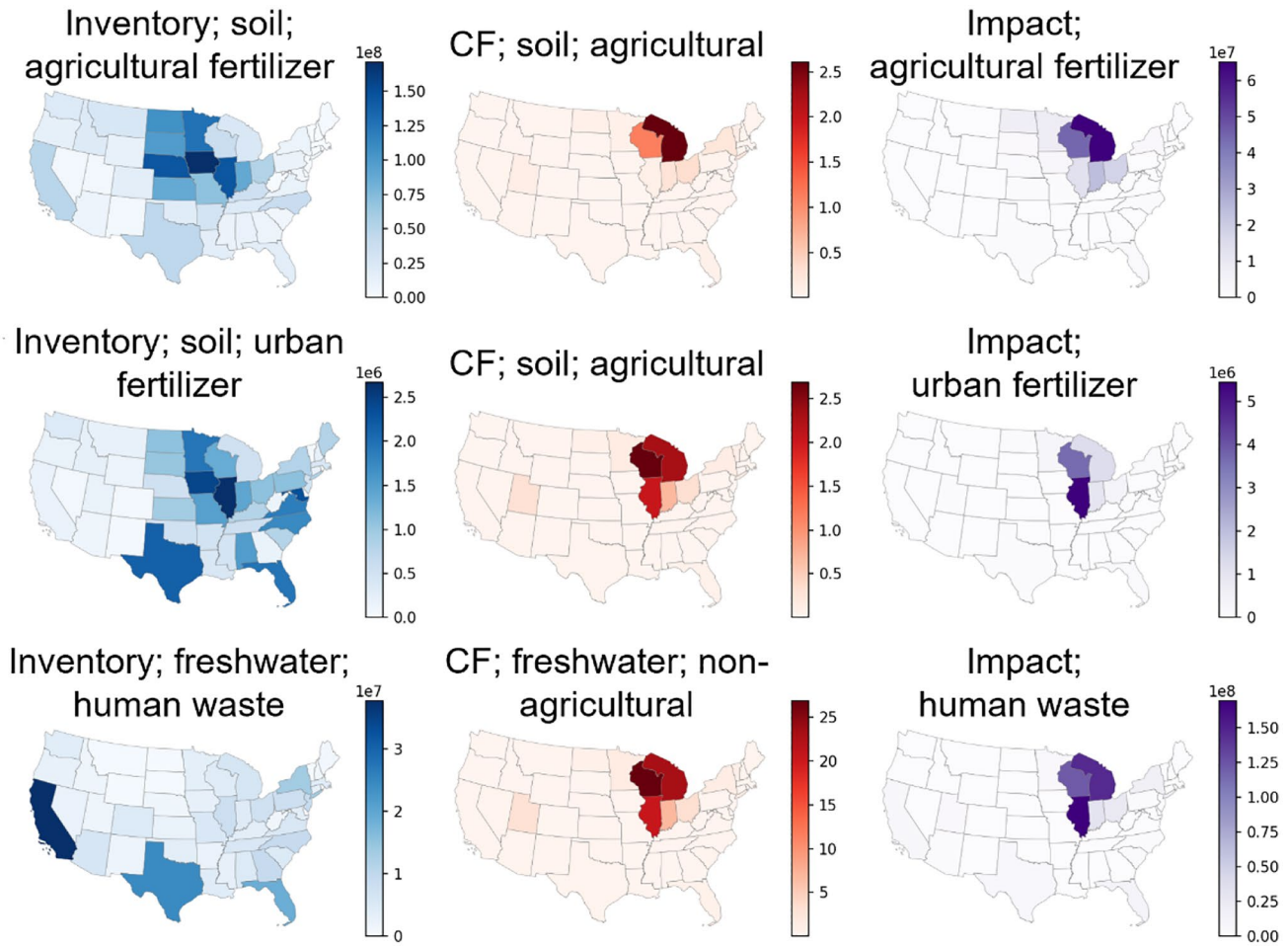

Fig. 7 Application case for N at the state level: agricultural fertilizer (top row), urban fertilizer (middle), and human N waste (bottom). Columns represent the national inventory at the state level (left column, kg N / yr), the corresponding characterization factor (center column, CF, yr), and the total impact (right column, kg N)

The marine eutrophication model for emissions to freshwater, soil, and LME shows variations across watersheds of up to a factor of ten. In contrast, CFs for NH_3 emissions show less drastic variations, for example, whereas NO_x emissions may vary by 20% from cell to adjacent cell. Overall, there is greater variation among state values for freshwater eutrophication than for marine eutrophication emissions from soil, to freshwater, or to LMEs. Both of these groups exhibit greater variation than marine eutrophication CFs from airborne emissions.

The residence time also affects the spatial variation of fate factors and CFs. The freshwater fate factor for phosphorus varies proportionally with the freshwater residence time. According to the underlying freshwater model, greater hydraulic residence time (especially in lakes where the phosphorus may not settle out of the water column) leads to a larger steady-state mass of phosphorus in the system.

Aggregation by likelihood can strongly influence an overall value. However, the variation is typically modest, due to relative sizes of the fate cells and aggregation targets, as well as the inherent fate variation, as previously discussed. Table 3 describes the variation at the state- and country-level.

As seen in Table 2 and Fig. 8, differences between likelihood categories influence the resulting characterization factors. For example, Illinois (Fig. 8) shows a 15-fold difference between the agricultural and nonagricultural CFs for freshwater eutrophication (see also top right panel of Fig. 5). The higher CF values associated with the relatively dense population in the Chicago area, compared with the rest of the state, contribute to a nonagricultural CF that is 15.4 times greater than the agricultural CF for the entire state of Illinois. For completeness, all relevant likelihood factors are included for all impact categories and spatial resolutions. Users of TRACI who know the

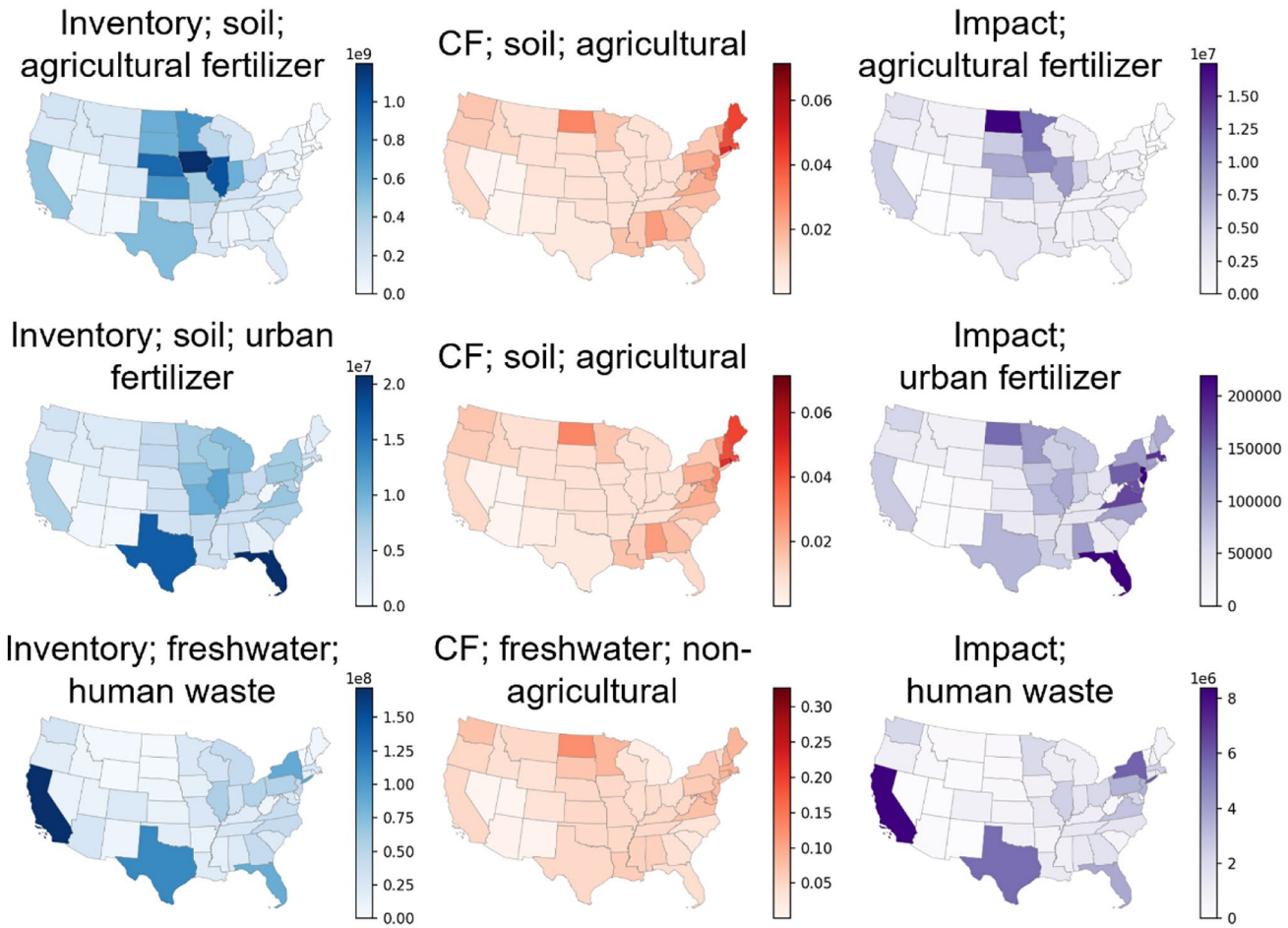

Fig. 8 Freshwater eutrophication grid for Illinois. Chicago, a population center, is in upper northeastern Illinois. This section of the state is also in the Great Lakes watershed, as seen by the higher (darker) CF values, whereas the remainder of the state is in the Mississippi watershed (with lower, i.e. lighter, CF values)

nature and location of an emission may select the appropriate activity likelihood and location.

3.1 Freshwater fate factors and the Great Lakes

As the Great Lakes are a unique system in the USA and the world, the effect of large lakes on the freshwater eutrophication factor deserves some mention. As noted in Fig. 8, there can be sharp gradients in freshwater fate factor from one location to the next, as one moves across watershed boundaries. The following discussion addresses the specific examples from the Great Lakes, but these trends can be observed anywhere there is a lake with a hydraulic residence time significantly larger than nearby systems.

Even among the Great Lakes, there can be a significant difference in hydraulic residence time, and thus the freshwater fate factor. The hydraulic residence time of the Great Lakes ranges from centuries (Superior: 191 years, Michigan: 99) to decades (Huron: 22) to a handful of years (Erie: 2.6, Ontario: 6) (US EPA 2015). These high hydraulic residence times have a noticeable impact on characterization factors as can be seen for characterization factors for states that drain into the Great Lakes (and specifically Lake Superior and Lake Michigan) in Fig. 4.

Therefore, the native (non-aggregated) freshwater fate factors for emissions upstream of Lake Superior or Lake Michigan will be a factor of 20–120 or more greater than the factors for Lake Erie or Lake Ontario. At the aggregated state (and county) level, the likelihood data also play a role. For example, although the Laurentian Watershed of northeast Minnesota drains into Lake Superior, there is relatively little agricultural activity or population in the watershed. Therefore, the state value for Minnesota is more heavily influenced by the freshwater fate factor for the Mississippi River basin (in which native factors are on the order of < 0.1 yr). The state of Michigan, however, has areas that drain into Lake Superior and Lake Michigan, and

Table 2 Eutrophication impact categories, likelihoods, and geographic aggregation targets, with associated characterization factors summarized across the interdecile range (IDR). Agric = agricultural; Nonagric = non-agricultural; IDR = the difference between the first and ninth deciles, i.e., 10th and 90th percentiles

Eutrophication category	Emission compartment	Flow	Likelihood	Aggregation target	10th percentile (yr)	50th percentile (yr)	90th percentile (yr)	IDR/median
Freshwater	Freshwater	P	Agric	State	0.045	0.15	1.2	7.3
Freshwater	Freshwater	P	Agric	County	0.0071	0.12	0.61	5.2
Freshwater	Freshwater	P	General	State	0.045	0.15	1.3	7.9
Freshwater	Freshwater	P	General	County	0.0071	0.12	0.6	5.2
Freshwater	Freshwater	P	Nonagric	State	0.045	0.16	2.9	18
Freshwater	Freshwater	P	Nonagric	County	0.0073	0.12	0.63	5.4
Marine	Freshwater	N	Nonagric	State	0.025	0.05	0.088	1.3
Marine	Freshwater	N	Nonagric	County	0.019	0.05	0.082	1.3
Marine	Soil	N	Agric	State	0.0075	0.012	0.041	2.9
Marine	Soil	N	Agric	County	0.0059	0.0083	0.028	2.7
Marine	LME	N	Nonagric	State	0.12	0.19	0.2	0.44
Marine	LME	N	Nonagric	County	0.073	0.19	0.19	0.63
Marine	Air	NH3	Agric	State	0.07	0.095	0.16	0.96
Marine	Air	NH3	Agric	County	0.071	0.094	0.16	0.89
Marine	Air	NH3	General	State	0.07	0.096	0.16	0.95
Marine	Air	NH3	General	County	0.071	0.094	0.16	0.89
Marine	Air	NH3	Nonagric	State	0.07	0.097	0.16	0.89
Marine	Air	NH3	Nonagric	County	0.071	0.094	0.16	0.89
Marine	Air	NOx	Agric	State	0.071	0.094	0.15	0.8
Marine	Air	NOx	Agric	County	0.068	0.09	0.13	0.65
Marine	Air	NOx	General	State	0.07	0.092	0.15	0.83
Marine	Air	NOx	General	County	0.068	0.09	0.13	0.64
Marine	Air	NOx	Nonagric	State	0.069	0.09	0.15	0.85
Marine	Air	NOx	Nonagric	County	0.068	0.09	0.13	0.65

the latter also has agricultural activity and high population density. Therefore, the overall state value for Michigan directly reflects the high freshwater fate factors for Lake Michigan. As a final example, Ohio has dense population and agricultural activity in both the Great Lakes basin and in the Mississippi River basin. Since the fate factors for Lake Erie and Lake Ontario are at least a factor of 20 smaller than the fate factors for the upstream Great Lakes, the state value for Ohio (in the Lake Erie watershed) is modest relative to Wisconsin or Michigan (in the upstream lake watersheds).

3.2 LMEs

Marine characterization factors are shown in Fig. 6. Note that the color gradient is variable for each map, and that

Table 3 Statistical variation across the CONUS aggregated characterization factors, using agricultural, nonagricultural, and general likelihoods for state-level and country-level geographic targets, as appropriate. Variation is calculated as the difference of the maximum and minimum value (i.e., 0 for CONUS), divided by the median. Note that variation across the following categories could not be calculated, as only two of the three (agricultural and nonagricultural) likelihood categories were included: marine eutrophication; N to soil, N to freshwater, and N to LME

Impact category and substance	Lower 48 states: max variation	Lower 48 states: median variation	Lower 48 country: total variation
Freshwater eutrophication; P to freshwater	15.4	0.21	1.7
Marine eutrophication; NH3 to air	0.15	0.02	0.23
Marine eutrophication; NOx to air	0.68	0.20	0.22

CFs are highly variable when comparing individual maps within the figure.

In particular, the characterization factors for emissions of nitrogen directly to marine systems are shown in the center column of the top row. There is a 50-km buffer around each LME, based on the assumption that only emissions within 50 km would be considered direct emissions to a large marine ecosystem. States located outside of this buffer therefore have no fate factor; only states within the 50-km buffer are included. The buffer distance, however, introduces the possibility that a state like Arizona, for example, could have an emission directly to the Gulf of California. The Gulf of California is a relatively enclosed LME, as compared to LMEs off the Atlantic or Pacific coasts. As such, the LME fate factor for the Gulf of California (0.65 yr) is three times greater than the coastal LME of the California Current (0.19 yr)., Therefore, the state CF value for Arizona is correspondingly high.

In the bottom row of Fig. 6, air emissions which deposit and influence the Hudson Bay watershed result in higher characterization factors for marine eutrophication than most other air emissions. CFs are also high along the Atlantic coast. These trends are driven by the atmospheric fate modeling of Roy et al. (2012) and the LME fate model of Cosme et al. (2017b). First, atmospheric transport in the USA generally occurs from west to east. All else being equal, a lower fraction of emissions on the West Coast will deposit in the Pacific LMEs than will emissions on the East Coast deposit on Atlantic LMEs. Secondly, the LME fate factor for the Hudson Bay is higher (1.4 yr) than for other LMEs in North America (the California Current, off the US West coast = 0.19 year, the Gulf of Mexico = 0.16 year, the Southeast US Continental Shelf = 0.07 year, the Northeast US Continental Shelf = 0.19 year, and the Newfoundland-Labrador Shelf = 0.19 year). Thus, the upper Midwest and Northeast USA have the highest fractions of emissions depositing in an LME (the Hudson Bay) with a higher FF than other LMEs. The use of a fate factor for direct emissions to LMEs should be used with caution, especially in the case of a state-wide value. There should be good evidence to support the fact that the emission occurs on the coast (or within 50 km of the coast).

3.3 Application case: N and P inventories

To demonstrate the application of CFs in the USA, nitrogen and phosphorus emissions available from the US Environmental Protection Agency (EPA) (US EPA 2018, 2019) were combined in a series of maps which are discussed below. This dataset was chosen because they represent the most comprehensive publicly available N and P inventories for the continental US (CONUS).

For both Figs. 4 and 5, it is important to realize that the CFs are intensive factors (meaning that they are independent of land area), whereas the inventory and impact are both dependent upon land area. This may result in some states (e.g., Texas and California showing up as being highly impacted when in reality, on a land mass basis, they may not be as impacted as other states.

For illustration, the following categories were mapped: human waste, agricultural fertilizer, and urban fertilizer. While runoff from fertilizer is minimized, model calculations are difficult to automate since runoff depends on the slope of the land, the condition of the soil at the time of application, the rainfall after application, etc. Therefore, a proxy estimate of 10% runoff is incorporated into the characterization factors of soil application to represent the estimated fertilizer runoff to freshwater.

Figure 6 illustrates the application case of P at the state-level. The center column are the CFs which were previously described in Fig. 4. The P inventory in the leftmost column reflects heavy use of agricultural fertilizer, as expected for the agricultural regions highlighted. Urban fertilizer inventory was difficult to characterize for states with significant agricultural production since in many cases it is not known whether fertilizer purchased is used for agriculture or urban fertilizers, so the left column center figure may have a disproportionate amount of fertilizer attributed to urban applications. These states tend to only report agricultural fertilizer use and therefore rely on county population to scale fertilizer application on non-agricultural land (Gronberg and Spahr 2012). Human waste can be seen in the bottom left figure and is consistent with expectations of population totals in each of these states.

The right-hand column includes the inventory and CFs multiplied for total potential impact. This shows that the influence of the CFs was much more dominant than the variations in inventory per state and that the highest eutrophication would be expected to be resulting from emissions from states which feed into the two Great Lakes with the longest retention times (Lake Superior and Lake Michigan).

Figure 7 illustrates the application case of N at the state-level. The leftmost column (blue) shows the inventory mapping, which follows the expected patterns. For example, human waste is centered around areas of high urban density, while urban fertilizer is found in areas where more suburban practices of lawn fertilizer application, golf centers, and other landscaping trends are popular. Agricultural fertilizer, pesticide, and herbicide use are more prevalent in the Midwestern states, consistent with livestock and grain farmland. The middle column of Fig. 8 (red) shows the CFs developed in this research, corresponding to the N inventories (blue). The combination of the inventories and the CFs is shown

in the rightmost column (purple) and represents the total potential impact.

For agricultural and urban fertilizers, the largest impacts are dependent upon both the combination of high inventories and the higher level of CFs which are prevalent for emissions in the Midwest and Atlantic states. Human waste impacts are highest in areas of high population, including California, Texas, New York and a few other coastal states.

The application case presented here is meant to display the relative influence of CFs and inventories and their level of influence on a CONUS example. We did not attempt to capture all of the emissions of N or P, but simply demonstrated the use of the CFs developed earlier in the paper.

Although these results are specific to this application, a few states' CFs significantly dominated the calculation of the total potential eutrophication impact. As mentioned previously, the retention times of the Great Lakes were a significant factor in these results. Globally, any large water body (e.g., the African Great Lakes or the Baltic Sea) could similarly influence a country-level CF. The marine eutrophication CFs and the inventories provided a more balanced influence on the potential impact within the USA.

4 Discussion

This paper builds upon the eutrophication model review presented in Morelli et al. (2018) and its set of recommendations for improvement to the eutrophication potential category in LCA. In the current work, we present a preliminary effort to spatially differentiate freshwater and marine eutrophication characterization factors to improve the scientific robustness of the impact category. The freshwater eutrophication fate model was provided by Helmes et al. (2012) while input to the marine eutrophication was provided by Cosme et al. (2017b) with characterization factors derived from NEWS2. In keeping with standard LCIA practice, we used the assumption that freshwater systems are P-limited and marine systems are N-limited. This assumption should be updated in future work, as co-limitation can occur in marine and freshwater systems.

Fertilizer sources of N and P played a key role in the application study used to evaluate the CF spatial patterns across the country. While some of the eutrophication impacts could be tracked according to agricultural production and lawn-care practices, this was not the case everywhere and fertilizer inventory in some locations were scaled by community population (as was the case for non-agricultural fertilizer use in a predominantly agricultural region). Despite using the best available data for N and P fertilizer sources, grid-based spatial data necessarily mask finer-scale (i.e., sub-grid) variabilities. This means that other sources of N and P that are not represented in the aggregate grid cell could, in fact, influence eutrophication. The secondary population-based estimate noted here introduces uncertainty into the dataset; this uncertainty could be quantified in future efforts, assuming the fertilizer estimates are not improved directly.

In the previous research of Morelli et al. (2018), the authors suggested several improvements to the eutrophication categories, beginning with the separation of freshwater and marine models. Recommendations that would require further research include (1) refining the spatial and temporal resolutions of the fate and transport models; (2) further developing the residence time estimates for freshwater and marine systems; (3) integrating soil data, such as soil type, to better estimate application and run-off of land-applied nutrients; and (4) using hydrodynamic models to expand the marine coverage beyond 66 LMEs. Throughout these improvements, researchers must continue to optimize the increasing spatial data resolution with the practical usability of available environmental models.

In addition to this method, there is also a eutrophication method under development by the United Nations Environment Program (UNEP) (UNEP 2019; Payen et al. in preparation). The co-authors of this paper are involved in the UNEP effort in various capacities and thus have insight into the compatibility of the two methods. While the TRACI method focuses on the contiguous USA and uses data at a finer resolution for the USA while also incorporating global data to produce country-level CFs, the UNEP method offers global CFs at a country resolution, and it uses the same likelihood data as TRACI. In addition, TRACI stops at the midpoint along the cause-effect chain, whereas the UNEP method continues along the chain to the endpoint, thus introducing another level of uncertainty. Both methods are spatially explicit eutrophication methods.

5 Conclusions

In this research, we have developed site-specific freshwater and marine eutrophication CFs at a fine scale for the US and at a larger scale for the rest of the globe. We have also applied freshwater and marine eutrophication factors for the USA for specific examples of agricultural fertilizers, urban fertilizers, and human waste.

Additional research is needed to provide similarly resolved CFs for the entire globe, which would require an expansion and deepening of the underlying data and fate and transport models. Environmental models currently available at higher-level resolutions tend to lack compatibility or feasibility with the LCA framework, often containing more detailed data than LCA can currently incorporate. However, these models could also replace common assumptions, such as the 10% runoff currently assumed across eutrophication models. Other additions to data could allow for further spatial refinement as

well as temporal refinement, especially for agricultural crop cover or seasonal scenarios. Endpoint modeling could also be incorporated once the science has reached a sufficient level of sophistication. Enhanced computing capabilities will allow for even better eutrophication modeling in LCA and extend to other impact categories in the future.

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Declarations

Disclaimer The views expressed in this article are those of the authors and do not necessarily represent the views or policies of the U.S. Environmental Protection Agency. Mention of trade names or commercial products does not constitute endorsement or recommendation for use.

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